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Opiate modulation of separation-induced distress in non-human primates

Ned H. Kalin, Steven E. Shelton and Charles M. Barksdale

Department of Psychiatry, University of Wisconsin-Madison and Psychiatry Service, William S. Middleton Memorial Veterans Hospital, Madison, WI 53705 (U.S.A.)

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Infant rhesus monkeys respond to separation from their mothers with a dramatic increase in vocalizations and activation of autonomic and pituitary–adrenal systems. Using the mother–infant separation paradigm in rhesus monkeys, we focused on the role of opiate systems in modulating the behavioral and neuroendocrine consequences of a brief, naturally occurring stressor. In the first experiment, morphine 0.1 mg/kg significantly decreased separation-induced vocalizations without affecting activity levels. In the second experiment, naloxone 1.0 mg/kg increased distress vocalizations but lower doses had no effect. In the third experiment we blocked the effect of morphine 0.1 mg/kg with naloxone 0.1 mg/kg, a dose of naloxone that had no intrinsic effects of its own. This suggests that the reduction of separation-induced vocalizations by morphine is mediated by opiate receptors. The last experiment demonstrated that separation-induced increases in pituitary–adrenal hormones can also be modulated by opiate agonists and antagonists. These findings are consistent with work in non-primate species and support the hypothesis that opiate receptors are specifically involved in mediating separation-induced vocalizations and pituitary–adrenal activation in primates.

INTRODUCTION

Extensive work in rodents[7,12,13], chickens[18,20–23,32], and dogs[19] suggests that endogenous opiate systems are important in modulating the distress response of vertebrate infants separated from their mothers. Distress vocalizations are the most prominent aspect of this response. In infant rats and chicks, distress vocalizations are influenced by various classes of neuropharmacological agents. Morphine, however, appears to have the most potent effects, and these are naloxone-reversible[13,22]. Infant primates show a dramatic behavioral and physiological response when separated from their mothers[1,4,9,25,28]. It is characterized by autonomic[24] and pituitary–adrenal activation[2,5,11], increased vocalizations[5,6,24,28], and behavioral agitation[6,24,28]. The 'coo' is the most frequent vocalization emitted by infant rhesus monkeys after maternal separation[27,28]. It has been suggested that this is a distress vocalization which serves an adaptive role by signaling the location of the infant so its mother can retrieve it.

Work in non-human primates demonstrates a neuroanatomical substrate specific for distress vocalizations. Using electrical stimulation techniques with rhesus monkeys, Robinson showed that the anterior cingulate cortex is necessary for the induction of coo vocalizations but not other vocalizations such as bark and shriek[26]. Other work with rhesus monkeys suggests that the anterior cingulate cortex has a role in learned vocalizations[30,31]. In a recent review, MacLean suggested that the anterior cingulate cortex is essential for the maintenance of attachment bonds[16]. As endogenous opiate systems have been implicated

Correspondence: N.H. Kalin, Psychiatry Service, William S. Middleton Memorial Veterans Hospital, 2500 Overlook Terrace, Madison, WI 53705, U.S.A.

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in the separation response, it is of interest that the cingulate cortex has a high density of opiate receptors[15,33].

We have been using the mother–infant separation paradigm in rhesus monkeys to explore the mechanisms mediating the consequences of a brief, naturally occurring stressor in a primate species[10]. The studies presented here focus on the opiate system's role in the behavioral and neuroendocrine response to separation. In the first experiment we administered morphine, an opiate agonist, to infant monkeys undergoing brief separation from their mothers. Based on work in other species, we anticipated that this would decrease separation distress. In the second experiment we administered the opiate antagonist naloxone. The purpose of this experiment was to characterize any intrinsic effects of naloxone on the separation response and to establish a dose of naloxone that could be used in the next experiment to block the effects of morphine. We hypothesized that naloxone either would have no effect or would increase separation-induced distress. The third experiment was designed to test whether the distress-reducing effects of morphine were specifically mediated through opiate receptors and to establish the lowest dose of naloxone capable of blocking morphine's effects on the separation response. The fourth experiment was aimed at establishing the neuroendocrine effects of brief separation and the role that opiate systems might play in separation-induced neuroendocrine activation.

MATERIALS AND METHODS

Subjects

The experimental subjects were 8 infant rhesus monkeys (*Macaca mulatta*), 4 males and 4 females, housed at the University of Wisconsin Harlow Primate Laboratory. At the beginning of the studies they ranged in age from 5 to 8 weeks and weighed between 0.43 and 0.70 kg. Each infant was housed with its mother in a cage 71.5 cm wide by 71.5 cm high by 68 cm deep. Animals were maintained on a 12-h light–dark cycle (lights on from 06.00 to 18.00 h). Infants were breast-fed by their mothers and remained with them at all times except during the short experimental separation periods.

Experimental design

A within-subjects design, balanced for the order of drug administration, was used. In each study 8 animals were used, and all of them received every drug dose.

Drug dosage and testing procedures

In the first experiment, each animal received vehicle (0.9% saline) and morphine sulfate 0.1 and 1.0 mg/kg. All doses were administered in a volume of 0.1 ml/kg. The 3 treatments were separated by intervals of 1 week.

On each test day at 10.00 h, 4 infants were separated from their mothers and immediately injected intramuscularly with drug. They were then moved to another room and placed in separate compartments of a larger cage that had been quadrisected by mesh panels into sections 75 cm high by 75 cm wide by 60 cm deep. This arrangement allowed the infants to see, hear, and smell each other as well as to touch each other if they desired. Behavioral testing was always begun at 10.15 h, by a trained observer who was unaware of the treatment conditions. Tests consisted of two 5-min sessions with a 30-min interval between sessions. One hour after separation the infants were reunited with their mothers.

In the second experiment, the animals received saline vehicle and 3 different doses of naloxone hydrochloride (0.01, 0.1 and 1.0 mg/kg). Drug was administered intramuscularly, at 4-day intervals, in a volume of 1.0 ml/kg. Infants were separated from their mothers at 10.00 h, drug was administered, and behavioral testing began at 10.15 h.

The third experiment followed a 2-by-4 design. At 10.00 h the infants were separated from their mothers and immediately injected with saline vehicle or naloxone (0.001, 0.01, 0.1 mg/kg) in a volume of 1.0 ml/kg. Five minutes later each animal received a second injection consisting of either the saline vehicle or morphine (0.1 mg/kg) in a volume of 1.0 ml/kg. Thus each animal received all 8 treatments, with 1 week between treatments. Behavioral testing began at 10.15 h, 15 min after separation.

In the fourth experiment, we tested 4 drug combinations (vehicle–vehicle, vehicle–morphine 0.1 mg/kg, naloxone 0.1 mg/kg–vehicle, and naloxone 0.1 mg/kg–morphine 0.1 mg/kg). Drugs were administered to infants under two conditions (with their

mothers and separated from their mothers) and blood was collected for determination of adrenocorticotropin (ACTH), prolactin and cortisol concentrations. The timing of drug injections was the same as in Expt. 3. The experiment was performed over 8 weeks, drugs being administered at weekly intervals. In weeks 1, 3, 5 and 7 infants remained separated from their mothers after drug administration and were behaviorally tested. In weeks 2, 4, 6 and 8 they stayed with their mothers after drug administration and were not behaviorally tested. All infants were caught 60–75 min after drug administration, and within 8 min 2.0 ml of blood was collected by femoral venipuncture.

Behavioral scoring and analysis

The behavioral scoring and software system we used was developed at the Harlow Primate Laboratory[14]. It utilizes a syntactic encoding process that records the frequency, duration, coincidence and sequence of behaviors. Our present system uses 35 operationally defined behavioral scoring categories. The behaviors most frequently scored in infants undergoing separation are defined in Table I.

Sound spectrograms of the 3 types of vocalizations defined by our scoring system were made using the method of Snowden and Hodin[29]. The spectrograms were made under the direction of Charles Snowden with a Kay Sound Spectrograph (Model 6061B) using the 80- to 80,000-Hz frequency range and a narrow band filter. Behavioral and hormonal data were analyzed by a repeated measures ANOVA. For vocalizations, data were recorded as frequencies; for all other behaviors, as duration. Duncan's *t*-tests were used for specific post-hoc comparisons.

Sample preparation and radioimmunoassays

All samples were collected into tubes containing EDTA and were immediately placed on ice. Plasma was separated from blood by centrifugation at 3000 rpm at 4 °C for 10 min and frozen at –70 °C until assayed.

ACTH immunoreactivity was measured in unextracted plasma with a completely characterized antiserum (IgG-ACTH-1, IgG Corp., Nashville, TN)[17]. This highly specific antiserum has an ED_{10} of 0.35 pg/tube. Intra-assay and interassay coefficients of variability are 4.2 and 14.6%, respectively. Serial dilutions of monkey plasma result in parallel displacement of the radioactive tracer compared with standard $ACTH_{1-39}$ (ref. 8).

Plasma cortisol was measured with reagents obtained from New England Nuclear (Boston, MA). The sensitivity of the assay is 2 μg/dl. Intra-assay variability is 2.9% and interassay variability is 4.8%. Plasma prolactin was measured with reagents obtained from Diagnostic Products (Los Angeles, CA). The antiserum is highly specific for prolactin and does not significantly cross-react with growth hormone or other anterior pituitary hormones. The ED_{10} of the assay is 0.7 ng/tube and serial dilutions of monkey plasma resulted in parallel displacement of ^{125}I-labeled prolactin compared with the prolactin standard.

TABLE I

Behavioral categories most frequently scored in infant monkeys undergoing separation from their mothers

Category	*Description*
Environmental exploration	Any manual or oral exploration or manipulation of the physical surroundings (cage, pan, droppings, playing with chow, etc.). Includes cage shake.
Locomotion	Ambulation of one or more full steps at any speed. Includes such behavior as dropping from ceiling to floor or swinging cage shake. May be scored with other behaviors.
Vocalizations	Any audible communication. 'Coo' is most frequent vocalization occurring with separation and is longest in duration. Produced by rounding and pursing the lips; characterized by first increase, then decrease in frequency and intensity. 'Bark' is used as threat or warning. Mouth is slightly open with lips in normal position; abdomen forces air through vocal cords, producing short, rasping, low-frequency sound. 'Shriek' is a harsh sound usually produced in the context of perceived immediate danger. An abrupt, loud, high-pitched sound with little change in frequency. Produced with lips pulled back exposing teeth.
Inactivity	Maintenance of stable, stationary position with no simultaneous scorable behavior except proximity and vocalization.

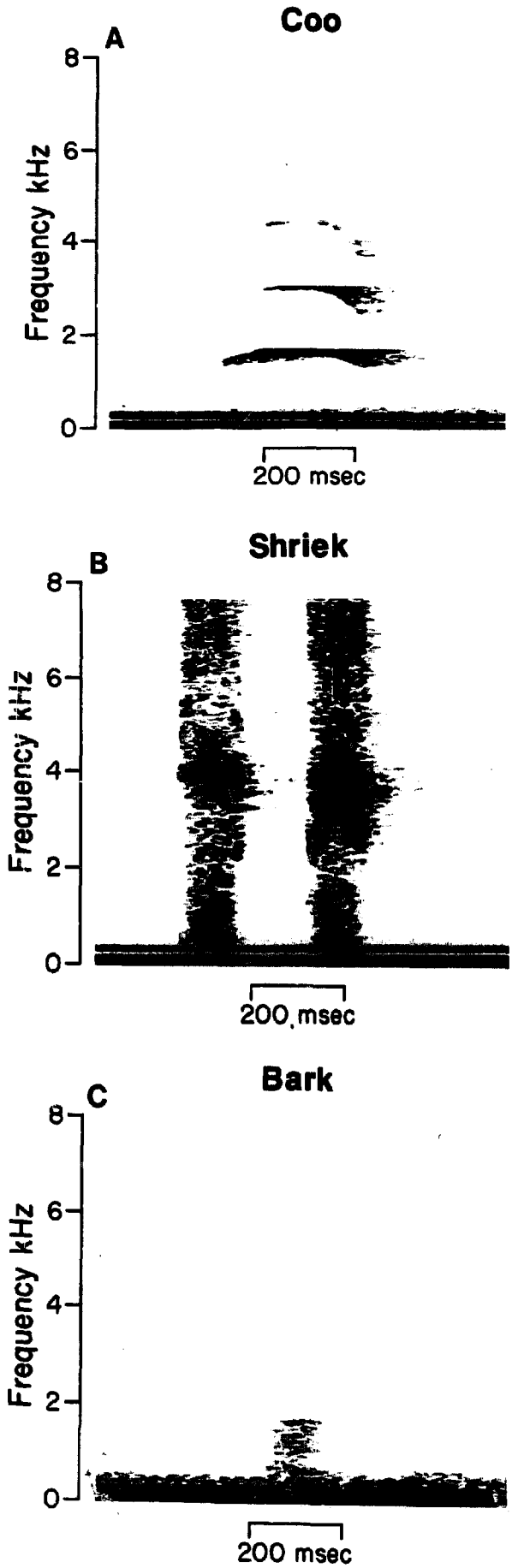


Fig. 1. Sonograms of (A) a coo, (B) a shrieks, and (C) a bark from an infant rhesus monkey separated from its mother. These vocalizations are described in Table I.

←

RESULTS

In all 4 experiments, animals that received the saline vehicle displayed an intense separation response characterized by frequent coo vocalizations and behavioral agitation. Fig. 1 displays typical spectrograms of a coo, a bark, and a shriek from the infants in this study*. The 'coo' is the most frequent vocalization that occurs with separation. Other common vocalizations are the 'bark' and the 'shriek' (see descriptions in Table I).

In the first experiment, morphine significantly af-

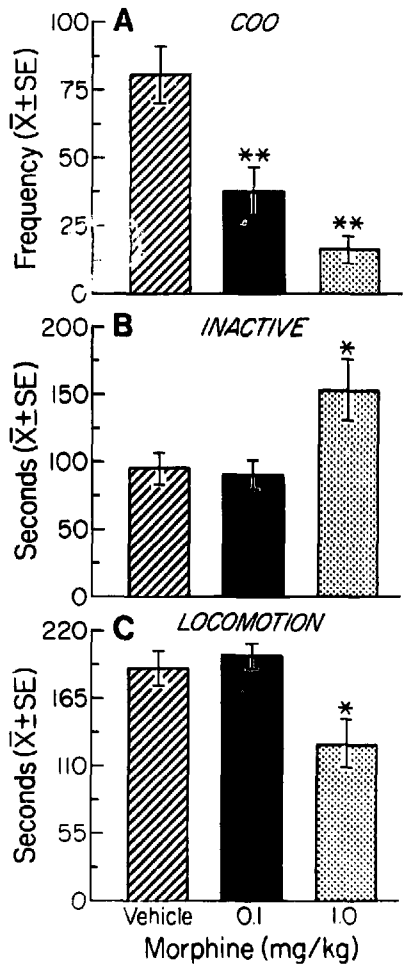


Fig. 2. Effects of morphine on separation-induced behaviors (coo vocalizations, inactivity, locomotion) in infant rhesus monkeys. $n = 8$; * $P < 0.05$; ** $P < 0.01$.

* Copies of the sonograms and an audiotape of the infant rhesus monkey vocalizations are available on loan from the Audiovisual Collection, Primate Center Library, Wisconsin Regional Primate Center, Madison, WI.

fected separation-induced vocalizations, locomotion and inactivity (Fig. 2). Repeated-measures ANOVA revealed a significant effect of dose on coo ($F_{2,14}$ = 14.70; $P < 0.0004$). After vehicle, the animals emitted a mean ± S.E.M. of 80.4 ± 14.3 coos per 5-min test session. This decreased to 37.6 ± 12.3 and 16.1 ± 6.1 after 0.1 and 1.0 mg/kg morphine. Both doses of morphine reduced coos ($P < 0.01$). Dose had a significant effect on locomotion ($F_{2,14}$ = 4.996; $P < 0.025$). Compared with vehicle, the 0.1 mg/kg dose of morphine did not alter locomotion; but morphine 1.0 mg/kg reduced locomotion in comparison with vehicle and 0.1 mg/kg morphine ($P < 0.05$). The effects of morphine on inactivity followed a similar pattern ($F_{2,14}$ = 4.191; $P < 0.04$): results with vehicle and morphine 0.1 mg/kg did not differ significantly, but 1 mg/kg morphine increased inactivity ($P < 0.05$). In summary, 0.1 mg/kg morphine markedly reduced separation-induced vocalizations without affecting the infants' activity level.

In the second experiment, naloxone affected separation-induced coo vocalizations and environmental exploration. Repeated-measures ANOVA revealed an overall effect of naloxone on coos ($F_{3,21}$ = 3.239; $P < 0.05$). Post-hoc tests showed significant effects only after the highest dose of naloxone, 1.0 mg/kg ($P < 0.05$). At this dose a mean of 90.70 ± 11.1 coos occurred, compared with 73.9 ± 11.1 with vehicle (Fig. 3A). An overall effect of naloxone was also found for environmental exploration ($F_{3,21}$ = 3.102; $P < 0.05$). Again, this behavior increased significantly only after the highest dose of naloxone (Fig. 3B).

The aim of the third experiment was to test whether morphine's effects on the separation response were naloxone-reversible and to identify the lowest dose of naloxone capable of reversing these effects. Data were analyzed using repeated-measures ANOVAs with 3 factors: (1) morphine (0.1 mg/kg and vehicle); (2) naloxone (0.001, 0.01, and 0.1 mg/kg and vehicle); and (3) test session.

Morphine reduced separation-induced coos ($F_{1,7}$ = 51.372; $P < 0.0002$). A main effect of naloxone was not found. However, there was a morphine × naloxone interaction ($F_{3,21}$ = 3.015; $P < 0.05$). As illustrated in Fig. 4, 0.1 mg/kg morphine administered with vehicle markedly reduced coos compared with vehicle–vehicle ($P < 0.01$). The effect of morphine was reversed only with the 0.1 mg/kg dose of naloxone. This drug combination resulted in a level of separation-induced coos that was significantly greater than when morphine was administered with vehicle ($P < 0.05$) and not different from that in the vehicle–

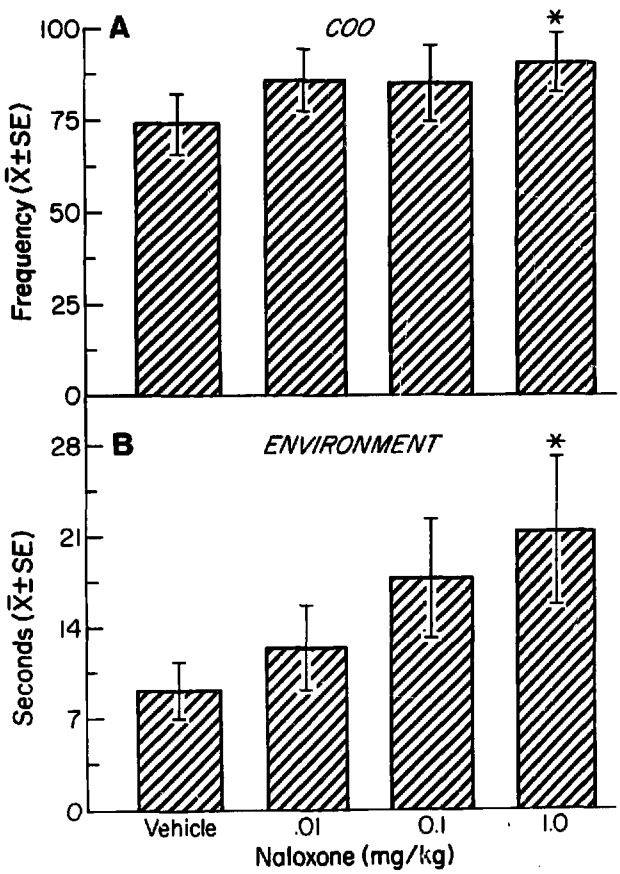


Fig. 3. Effects of naloxone on separation-induced coos and environmental exploration. n = 8; * $P < 0.05$.

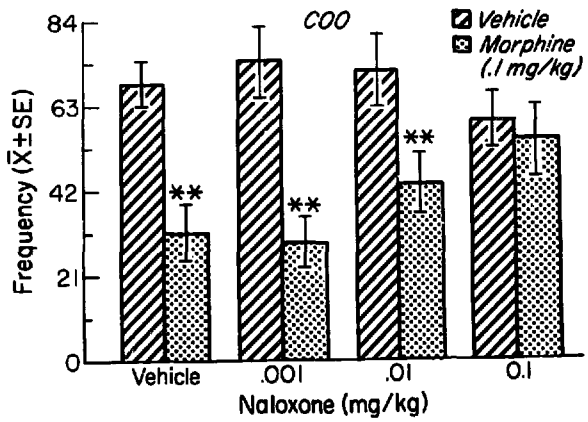


Fig. 4. Effects of naloxone on reduction by morphine of separation-induced coos. n = 8. ** Indicates significant difference ($P < 0.01$) between morphine treatment and vehicle at the same dosage of naloxone.

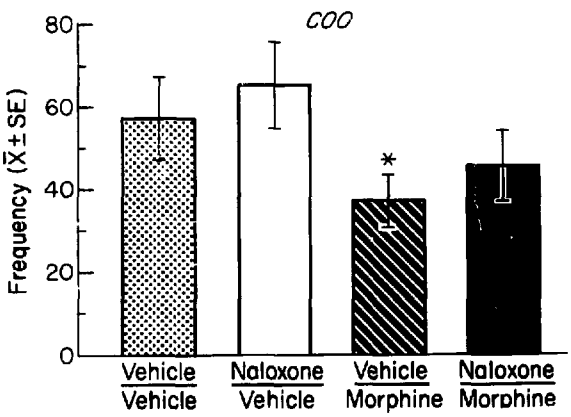


Fig. 5. Effects of combinations of vehicle, naloxone and morphine on distress vocalizations. $n = 8$; * $P < 0.05$.

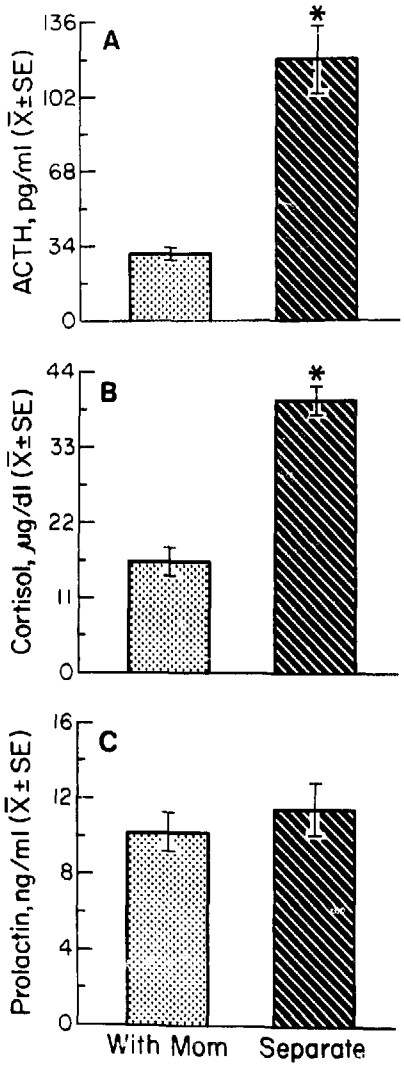


Fig. 6. Plasma concentrations of (A) ACTH, (B) cortisol, and (C) prolactin in 8 infant rhesus monkeys with (stippled bars) and separated from (diagonally striped bars) their mothers. * $P < 0.005$.

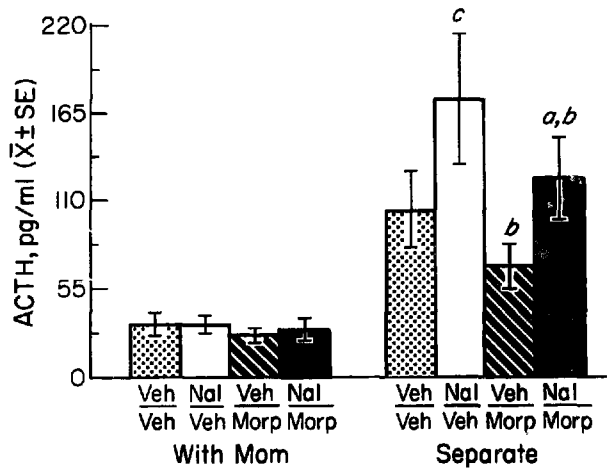


Fig. 7. Effects of vehicle (Veh), morphine (Morp), and naloxone (Nal) combinations on plasma ACTH concentrations in 8 infant rhesus monkeys when with and separated from their mothers. a, differs from Veh/Morp (Separate), $P < 0.05$; b, differs from Nal/Veh (Separate), $P < 0.05$; c, differs from Veh/Veh (Separate), $P < 0.01$.

vehicle combination. The only other behavior affected by the drug treatments was environmental exploration. A main effect was found for naloxone ($F_{3,21} = 5.712$; $P < 0.005$) but not for morphine. Only the 0.1 mg/kg dose of naloxone resulted in significantly more environmental exploration than vehicle–vehicle (21.2 ± 6.7 compared with 8.8 ± 3.6 s).

The fourth experiment assessed the effects of separation on neuroendocrine function and tested whether activation of opiate systems would attenuate these effects. Based on the results of the third experiment, naloxone 0.1 mg/kg was used to antagonize morphine 0.1 mg/kg. Fig. 5 demonstrates the overall effect of the drug treatments on coos ($F_{3,21} = 4.228$; $P < 0.02$). As in the preceding experiments, morphine 0.1 mg/kg reduced separation-induced coos ($P < 0.05$) without affecting locomotor behavior. Naloxone 0.1 mg/kg did not affect coos and partially blocked the effects of morphine, as indicated by the lack of significant difference between the morphine–naloxone and vehicle–vehicle drug combinations. This effect of naloxone 0.1 mg/kg was less prominent than that seen in Expt. 3.

With the vehicle–vehicle treatment, separation from the mother resulted in marked release of ACTH ($P < 0.005$) and cortisol ($P < 0.005$) but not prolactin (Fig. 6). Fig. 7 compares the effects of the various treatments on ACTH concentration when the infants were with their mothers and separated from them. The ANOVA revealed an effect on

ACTH of housing condition (with mother or separated) ($F_{1,7} = 18.479$; $P < 0.004$) and drug treatment ($F_{3,21} = 5$; $P < 0.007$), and a housing condition × drug treatment interaction ($F_{3,21} = 5.305$; $P < 0.02$). Although none of the drug treatments significantly altered ACTH concentrations when the infants were with their mothers, drug effects were found when the infants were separated. In infants separated from their mothers, the morphine–vehicle treatment resulted in non-significantly lower ACTH concentrations than the vehicle–vehicle treatment (68.5 ± 13.8 compared to 103.8 ± 22.7 pg/ml). The naloxone–morphine combination reversed the effects of vehicle–morphine, resulting in ACTH levels of 124.0 ± 26.4 pg/ml ($P < 0.05$). Naloxone–vehicle increased ACTH compared to vehicle–vehicle ($P < 0.01$).

Cortisol concentrations also were affected by housing condition ($F_{1,7} = 143.90$; $P < 0.0001$) and drug treatment ($F_{3,21} = 6.007$; $P < 0.004$). However, there was no significant interaction between housing condition and drug treatment. The vehicle–morphine treatment reduced cortisol compared with vehicle–vehicle ($P < 0.05$) and this effect was significantly blocked by the prior administration of naloxone. Neither housing condition nor drug treatment affected concentrations of prolactin.

DISCUSSION

Studies in non-primate species have demonstrated that opiate systems modulate the effects of maternal–infant separation. Our results extend these findings to primates, showing that exogenously administered opiates reduce the behavioral and neuroendocrine effects of attachment bond disruption in infant monkeys. Our finding that naloxone increases separation-induced distress vocalizations and ACTH suggests an involvement of endogenous opiate systems in the primate's response to separation. This is further supported by our observation that naloxone administered to infant rhesus monkeys that remain with their mothers does not elicit vocalizations (unpublished data).

In earlier work with infant monkeys, we showed that benzodiazepines reduced separation-induced vocalizations and pituitary–adrenal activation and increased activity[10]. We were unable to reverse the effects of diazepam on vocalizations with the benzodiazepine antagonist Ro 15-1788. However, Ro 15-1788 did reverse the effects of diazepam on activity and separation-induced release of ACTH and cortisol. These findings suggest that effects of diazepam on stress-induced activity changes and pituitary–adrenal activation, but not on vocalizations, are specifically mediated by benzodiazepine receptors. The present studies support a specific involvement of opiate receptors in mediating separation-induced vocalizations and pituitary–adrenal activation. Since others have demonstrated benzodiazepine–opiate interactions[3,34], it is possible that the effects of benzodiazepines on vocalizations are mediated through opiate receptors. Our findings suggest that the neuroendocrine and behavioral consequences of attachment bond disruption in primates are mediated by both opiate and benzodiazepine systems.

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